



# United International University

## School of Science and Engineering

Mid Term Examination; Year 2026; Trimester: Spring

Course: PHY 2105; Title: Physics; Sec: All

Full Marks: 30, Time: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled as per UIU disciplinary rules.

**Questions no 1, 2, 3, 4 and 5 are mandatory to answer.**

- 1.(a) Plot the following equations of displacement vs. time for two different oscillations with 2 CO1  
equations,  $y = A \sin \omega_0 t$  and  $y = A \sin \left( \omega_0 t + \frac{\pi}{4} \right)$ . What is the phase difference between  
the two particles?

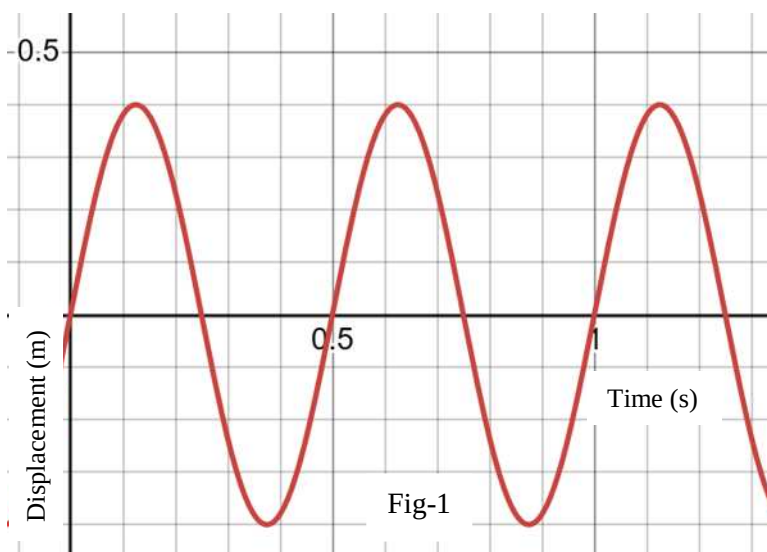
- (b) Robert connected a resistor in an LC circuit and the current flow decreased to 0 after some 2 CO1  
time. Why does it happen?

2. A mass spring system is pulled up to 5 cm from equilibrium and released at  $t = 0$ . The 3 CO3  
system completes 20 oscillations in 2 seconds. The mass of the system is 0.5 kg. Calculate  
the following:

- (i) Angular frequency of the oscillation  
(ii) Total energy of the system.

3. Fig-1 shows a Simple harmonic motion (SHM) as a 6 CO3  
function of time. The spring constant of the mass spring  
system is 19.6 N/m. Calculate  
the following terms:

- (i) equation of displacement  
(ii) velocity of the system at  $t = 0.9$  s  
(iii) kinetic energy at  $t = 0.9$  s.



4. The spring constant, mass, and damping coefficient of a mass-spring system in a damped 6 CO3  
harmonic oscillation (DHO) are  $k = 196$  N/m,  $m = 500$  g, and  $b = 40$   $\text{gs}^{-1}$ , respectively. The  
system was pulled up to 40 cm from the equilibrium and released at  $t = 0$ .

- (i) What will be the amplitude at 0.5 sec?  
(ii) What will be the displacement at  $t = 0.5$  sec?  
(iii) What will be the damping energy of the DHO at  $t = 0.2$  se

5. The capacitance, resistance and inductance of an LRC circuit is  $C = 10$   $\mu\text{F}$ ,  $R = 20$   $\Omega$ , and 3 CO3  
 $L = 5$  mH.

- (i) Determine whether the circuit is oscillatory or not?  
(ii) Determine the frequency of oscillations, if any.  
(iii) If we replace the resistor with different one, how much resistance is needed for the  
circuit to be critically damped?

**Answer any one from 6 and 7**

- 6.(a) The equation of displacement of a mass-spring system in a SHM is,  $y = A \sin \omega_0 t$ . Derive the equations for velocity, acceleration, potential energy, kinetic energy, total energy of the system. 5 CO2
- (b) Using the above equation ( $y = A \sin \omega_0 t$ ) plot a graph of energy vs. time for the mass-spring system and indicate (using arrow) on figure the kinetic energy, potential energy, and total energy. 3 CO2

- 7.(a) The differential equation of DHO in a mass spring system is,  $\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega_0^2 x = 0$ . Show that the solution of the equation is, 5 CO2

$$x = A e^{-\frac{b}{2m}t} \cos(\omega_d t + \varphi)$$

- (b) Derive the differential equation for an LC circuit which produces SHM and also express the parameters  $\omega$ ,  $T$ , and  $f$  of the oscillations. 3 CO2

**CO1:** Define different physical quantities with examples **CO2:** Derive/Show the various equations of SHM, DHM, wave motion etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of SHM, DHM.